

MIHIR KHATRI

Undergraduate Researcher in AI & Machine Learning

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PROFESSIONAL SUMMARY

Undergraduate researcher (GPA 9.4/10.0) specializing in **LLM inference efficiency**, alignment-adjacent interpretability, and GPU-aware compression architectures. Proven background in rigorous statistical hypothesis testing, multi-subset evaluation, and building robust end-to-end ML pipelines. Authored an independent first-author research preprint (Zenodo) on quantisation-compatible KV-cache compression evaluated across leading open-weight transformer models.

KEY RESEARCH & PROJECTS

- EEG-LLM Decoding System (Independent Research)** 2025
 - Designed an Elastic Net neural decoder integrated with SMOTE oversampling, robust scaling, and cross-subject domain adaptation.
 - Engineered temporal-window feature extraction routines (200-800ms post-stimulus) targeting cognitive ERP components.
 - Achieved >85% classification accuracy on neural pattern decoding across complex benchmark EEG datasets.
- Transformer Encoder-Decoder from Scratch** 2025
 - Implemented full multi-head self-attention, causal autoregressive masking, sinusoidal positional encoding, and layer normalization in PyTorch.
 - Conducted deep numerical analyses of gradient flow, vanishing activations, and training stability under varied initialization regimens.
 - Benchmarked convergence speed and BLEU/ROUGE metrics against standard BERT and T5 baselines on text summarization.
- Quantisation-Compatible KV-Cache Compression (Preprint)** 2025
 - Developed a novel selective pruning and compression algorithm preserving attention saliency in low-bit quantized environments.
 - Attained 60% KV-cache memory reduction with less than 1% degradation in perplexity and downstream task accuracy.
 - Evaluated empirically across Llama-2-7B and MPT-7B models under INT4, INT8, and FP16 quantization schemes.

EDUCATION

Bachelor of Technology in Computer Science and Engineering 2022 - 2026
Savitribai Phule Pune University, Maharashtra, India | **CGPA:** 9.4 / 10.0 | **Research GPA:** 9.4 / 10.0

TECHNICAL SKILLS

Deep Learning & LLMs:	PyTorch, Hugging Face Transformers, FlashAttention, KV-cache optimisation, Quantisation (AWQ/GPTQ), vLLM
Machine Learning:	Elastic Net, SMOTE, Scikit-learn, XGBoost, Feature Selection, Statistical Significance Testing
Tools & Infrastructure:	Git, Docker, Linux, Bash, Jupyter, Pandas, NumPy, Matplotlib, ReportLab, python-pptx
Research Methods:	Preprint Authoring, Paper Replication, Empirical Evaluation, Ablation Studies, Reproducibility

PUBLICATIONS & PREPRINTS

"Quantisation-Compatible KV-Cache Compression for Efficient LLM Inference"
First-Author Independent Preprint | Published on Zenodo (2025) | Open-source benchmarks and evaluation code on Llama and MPT architectures.